

Dev Jariwala

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EDUCATION

University of Michigan

B.S.E. in Computer Science

Ann Arbor, MI

May 2028

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Object-Oriented Programming

EXPERIENCE

Hyundai America Technical Center (HATCI) via KPIT Technologies

Novi, MI

Software Engineer Intern, Client Assignment

July 2026 – Aug. 2026

- Automated **20+** tests in Python with Robot Framework, cutting regression time **93%** from **6 hours** to **25 minutes**
- Expanded SiriusXM 360L regression coverage across **10** vehicle trims, improving overall infotainment release validation
- Identified **15+** defects across **5** infotainment modules, improving software release quality and system level reliability
- Validated **10+** release fixes through manual in-vehicle UI testing, verifying defect resolutions across production builds

KPIT Technologies

Novi, MI

Software Engineer Intern

June 2026 – July 2026

- Built a JWT-based authentication platform in Angular, Flask, and MySQL, securing access across **10+** internal tools
- Eliminated shared credential risk for **50+** internal users by enforcing granular role based authorization through JWTs
- Cut new user onboarding from **1** day to under **10** minutes through building **6** self service authentication **REST APIs**

Necessities

Ann Arbor, MI

Software Engineer Intern

Sep. 2025 – Apr. 2026

- Engineered **50+** reusable React components powering marketplace search, filtering, and real-time product discovery
- Enabled messaging for **100+ active users** with **PostgreSQL**-backed REST APIs, driving **500** conversations/month
- Reduced page load times by **30%** across **1,000+** listings through targeted PostgreSQL query and index optimization
- Automated CI/CD pipelines to ship **8+ production** releases with a 5-person team, improving deployment reliability

PROJECTS

StormRoute | [GitHub](#) · [Live Demo](#) | *Next.js, TypeScript, React, Leaflet, Open-Meteo, OSRM, Vitest*

- Engineered a Next.js/TypeScript routing engine to score **6** departure windows and identify the safest departure time
- Integrated hourly forecasts with road geometry across **3** routes using Leaflet and OSRM, enabling route risk analysis
- Ranked 3 candidate routes using a deterministic **0–100** risk model combining **6** real-time weather conditions per route
- Cut repeat route analysis from **4.5s** to under **40ms** by caching weather forecasts keyed to individual route segments
- Validated the routing pipeline with **82** Vitest test cases, achieving **98.5%** line coverage and **94.0%** branch coverage

MiniRedis | [GitHub](#) | *C++20, TCP/IP, Multithreading, CMake*

- Built a Redis-compatible key-value store from scratch in **C++20**, serving **6** core commands over raw TCP connections
- Engineered concurrent request handling with reader-writer locks and a worker pool to safely parallelize all operations
- Preserved **250+** operations across system crashes using append-only logging, TTL expiration, and LRU cache eviction
- Sustained **52,000+** operations/sec across **25,000** benchmarked requests under sustained concurrent client workloads

Cricket Scoring & Analytics Platform | *React, TypeScript, Supabase, PostgreSQL, GCP*

- Synchronized **3+** live scoring workflows by streaming shared match state over WebSockets for real-time data updates
- Encoded **15+** tournament rules, including dismissals, into a unified scoring engine for automated match day processing
- Automated second-innings targets and live player statistics across **8 scoring workflows**, eliminating manual tracking
- Implemented **3** authorization layers using **OAuth 2.0**, row-level security, and **RBAC** to secure real-time data access

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, SQL

Frameworks/Libraries: React, Next.js, Angular, Flask, Node.js, Robot Framework, Flutter

Databases: PostgreSQL, MySQL, Firestore, Supabase

Cloud/Developer Tools: Google Cloud Platform (GCP), Docker, Git, Linux, Vercel, CMake, CI/CD